Chapter 6 Notes

Section 1

Notes

Note that $F \upharpoonright A$ does not necessarily equal $A \times F''(A)$. In fact, if $F''(A)$ contains more than one element (say it contains at least the two elements b, c), then obviously $F \upharpoonright A \neq A \times F''(A)$, as, for any element $x \in A$, we would have $\langle x, b\rangle$ and $\langle x, c\rangle$ both in $A \times F''(A)$, but there can only be one y for which $\langle x, y\rangle \in F \upharpoonright A$.

It is remarked that:"If ϕ_1 and ϕ_2 are functions from a class A onto a class B and if $\phi_1\phi_2^{-1}$ is an identity function, then obviously $\phi_1 = \phi_2$." Here is why this is. Now, note that this assumes that ϕ_2^{-1}, is, in fact, a function. Then, for every element y in the range of ϕ_2, we have $\phi_2^{-1}(y) = x$ for some x in the domain of ϕ_2, and $\phi_1\phi_2^{-1}(y) = \phi_1(x) = y$. But also, $\phi_2(x) = y$. Thus $\phi_1(x) = \phi_2(x)$ for every x in the domain of ϕ_2, and because they have the same domains, the functions are the same.

Note that if a class A is well ordered under $\subseteq$, then it is also properly well ordered under $\subseteq$. The proof of this is the same as the proof of Proposition 1.3.

Exercise 1.1

Let $\mathcal{N}$ be a nest of functions.

(1): We first show that $\cup\mathcal{N}$ is function. Obviously $\cup\mathcal{N}$ is a relation. Now, suppose we had $\langle x, y\rangle$ and $\langle x, z\rangle$ both in $\cup\mathcal{N}$. Thus, for some functions f and g, $\langle x, y\rangle \in f$ and $\langle x, z\rangle \in g$. But, because $\mathcal{N}$ is a nest, either $f \subseteq g$ or $g \subseteq f$. Without loss of generality, suppose $f \subseteq g$. Then we would have $g(x) = y$ and $g(x) = z$, and because g is a function, $y = z$. Hence $\cup\mathcal{N}$ must be a function.

(2): We now show that the domain of $\cup\mathcal{N}$ is the union of the class of domains of the elements of $\mathcal{N}$ (call this D). Suppose $x \in \mathrm{Dom}(\cup\mathcal{N})$. Then, for some y and some function $f \in \mathcal{N}$, $\langle x, y\rangle \in f$, meaning $x \in D$. On the other hand, suppose $x \in D$. Then, for some y and some function $f \in \mathcal{N}$, $\langle x, y\rangle \in f$, meaning $x \in \mathrm{Dom}(\cup\mathcal{N})$. Thus $\mathrm{Dom}(\cup\mathcal{N}) = D$. Similar reasoning shows $\mathrm{Ran}(\cup\mathcal{N})$ is the union of the class of ranges of the elements of $\mathcal{N}$.

(3): Lastly, suppose that every element of $\mathcal{N}$ was 1-1. Now, suppose x and y are in the domain of $\cup\mathcal{N}$, and that $x \neq y$, and for some v and w, $\langle x, v\rangle \in \cup\mathcal{N}$ and $\langle y, w\rangle \in \cup\mathcal{N}$. Then, for some functions f and g in $\mathcal{N}$, $\langle x, v\rangle \in f$ and $\langle y, w\rangle \in g$. Without loss of generality, suppose that $f \subseteq g$. Then g is 1-1, meaning $v \neq w$. Hence $\cup\mathcal{N}$ is 1-1.

Exercise 1.2

First, let us note that On is not a set. For if it were, it would then be a transitive set of ordinals, and thus would be an ordinal. Then that would mean $On \in On$, which never holds for ordinals. Thus On is not a set.

Now, let A be the class $On \cup \{\{\{\emptyset\}\}\}$. Thus $x \in A$ iff either x is an ordinal, or $x = \{\{\emptyset\}\}$. Note that $\{\{\emptyset\}\}$ is not an ordinal, as if it were, then it would be transitive, which would imply that $\emptyset \in \{\{\emptyset\}\}$, which is not the case. Hence On is a proper subclass of A.

Then define the relation $x \leq y$ (on A) as follows: $x \leq y$ iff x and y are both ordinals and $x \subseteq y$, or $y = \{\{\emptyset\}\}$. Note that this defintion implies that $\{\{\emptyset\}\} \leq y$ only holds if $y = \{\{\emptyset\}\}$. We first show that $\leq$ is a well ordering of A.

O_1: If x is an ordinal, then $x \subseteq x$, so that $x \leq x$ holds. If $x = \{\{\emptyset\}\}$, then $x \leq x$ by definition.

O_2: Suppose $(x \leq y) \wedge (y \leq z)$. If all of x, y and z are ordinals, then obviously $x \leq z$. If any of x, y and z were equal to $\{\{\emptyset\}\}$, then $z = \{\{\emptyset\}\}$ must hold, which of course implies that $x \leq z$.

O_3: Suppose $x \leq y$ and $y \leq x$. Then either $x = y = \{\{\emptyset\}\}$, or both x and y are ordinals, in which case $x \subseteq y$ and $y \subseteq x$, so that $x = y$. Thus $x = y$.

O_4: If x and y are both ordinals, then of course either $x \leq y$ or $y \leq x$. If $y = \{\{\emptyset\}\}$, then $x \leq y$ (similarly with x).

O_5: For any subclass B of A that only contains ordinals, then, because On is well ordered under $\subseteq$, it contains a least element. If $\{\{\emptyset\}\} \in B$, then $B - \{\{\{\emptyset\}\}\}$ contains a least element e, and e is also the least element of B itself, as $e < \{\{\emptyset\}\}$. Thus every subclass B of A contains a least element.

Thus $\leq$ is a well ordering of A. Now, On is a proper lower section of A: It is a proper subclass of A, and $x < \{\{\emptyset\}\}$ for all elements in On. But On is not a set. Thus we have an example of a well ordering which is not a proper well ordering.

Section 2

Notes

Note that any subclass of an isomorphism from A onto B is an isomorphism from a subclass of A onto a subclass of B.

Obviously, if A is isomorphic to B, then B is isomorphic to A (as seen in the next paragraph).

Note that, if ϕ is an isomorphism from A onto B, then ϕ^{-1} is an isomorphism from B onto A. For, clearly ϕ^{-1} is a 1-1 function from B onto A. And suppose ϕ is an isomorphism from the system $\langle A, \leq_1 \rangle$ to the system $\langle B, \leq_2 \rangle$. Then, for any a_1, a_2 in B, $a_1 <_2 a_2$ iff $\phi(a_1) <_1 \phi(a_2)$.

The above fact is used, for instance, in Corollary 2.2. Also, in Corollary 2.2, note that $(2')$ follows because $\phi(\phi^{-1}(a)) = a$.

I think there is an error in the proof of Corollary 2.3. It claims that $\phi_1\phi_2^{-1}$ is an isomorphism from A onto A, but this is clearly not the case. For ϕ_2^{-1} is from B onto A, and ϕ_1 is from A onto B, meaning $\phi_1\phi_2^{-1}$ is actually from B onto B (now, this would also work, as then, by Corollary 2.2, $\phi_1 = \phi_2$, but it is explicitly said in the proof that $\phi_1\phi_2^{-1}$ is from A onto A). If we wanted to get a function from A onto A, we would need to pick $\phi_2^{-1}\phi_1$ (for example; there are others).

Here is why $\phi_1\phi_2^{-1}$ is in fact an isomorphism. For, if a_1 precedes a_2 in B, then $\phi_2^{-1}(a_1)$ precedes $\phi_2^{-1}(a_2)$ in A, and $\phi_1(\phi_2^{-1}(a_1))$ precedes $\phi_1(\phi_2^{-1}(a_2))$ in B (more generally, if f is an isomorphism from A onto B, and g is an isomorphism from B onto A, then fg is an isomorphism from B onto B).

Also, here is why (as said in Corollary 2.3), if $\phi_1\phi_2^{-1}$ is the identity map (for ϕ_1 and ϕ_2 both from A onto B), then $\phi_1 = \phi_2$. For suppose that $\phi_1\phi_2^{-1}$ is the identity map. Then, for any $x \in B$, $\phi_2^{-1}(x) = y$ for some $y \in A$, and $\phi_1(y) = x$, but also $\phi_2(y) = x$, meaning $\phi_1 = \phi_2$.

Corollary 2.5 holds, as, for any two distinct ordinals α, β, they are comparable. Without loss of generality, suppose $\alpha < \beta$. Now, α must be an initial segment of β (clearly, all elements of α are in β, due to the transitivity of ordinals). This is because, for any element $x \in \alpha$, and for any element $y \in \beta$ but not in α, $x < y$ holds because $x < \alpha$ (by assumption), and, if $x \geq y$, then $y < \alpha$. But, this would mean $y \in \alpha$, a contradiction. Hence α is an initial segment of β, and thus there can be no isomorphism from β to α.

Corollary 2.6 holds, as, between two distinct lower sections of the same well ordered system, one is an initial segment of the other. For supppose the two lower sections were $L_<(x)$ and $L_<(y)$. Now, without loss of generality, suppose $x < y$

(and thus $L_<(x)$ is a subset of $L_<(y)$). Then $L_<(x)$ is an initial segment of $L_<(y)$, as, for every $z \in L_<(x)$, obviously $z < w$ for every $w \in L_<(y)$ but not in $L_<(x)$ (for if not, then $z \geq x$), meaning $L_<(x)$ is proper lower section of $L_<(y)$, and hence also an initial segment of $L_<(y)$. Thus, if two distinct lower sections of the same well ordered system are isomorphic to each other, then one of them would be isomorphic to one of its initial segments, contrary to Corollary 2.4.

If A is isomorphic to B, and B isomorphic to C, then A is obviously isomorphic to C. For suppose A is isomorphic to B under ϕ_1, and B is isomorphic to C under ϕ_2. Then let ϕ_3 be the class of all orderd pairs $\langle x, \phi_2(\phi_1(x))\rangle$ where $x \in A$; ϕ_3 is the isomorphism from A to C.

This explains Corollary 2.7: For suppose a lower section of A was isomorphic to two distinct lower sections C and D of B. Then it would follow that A is isomorphic to C, and A is isomorphic to D, meaning C is isomorphic to D. But this would contradict Corollary 2.6.

In Corollary 2.8, here is why $I_2 \upharpoonright D_1$ is an isomorphism from D_1 onto a lower section of B: Clearly $I_2 \upharpoonright D_1$ is an isomorphism from D_1 onto a subclass of the range of I_2. Also, suppose that A was well ordered under $\leq_1$ and B was well ordered under $\leq_2$. Now, suppose C is the range of I_2, and C' is the range of $I_2 \upharpoonright D_1$. Suppose that C' was not a lower section of C. Then, for some $x \in C'$, there would be some $y \in C - C'$ such that $y <_2 x$. But, because I_2 is an isomorphism, $I_2^{-1}(y) <_1 I_2^{-1}(x)$. But $I_2^{-1}(x) \in D_1$, and it was assumed that $D_1 \subseteq D_2$, meaning $I_2^{-1}(x) <_1 I_2^{-1}(y)$ (because D_1 and D_2 are both lower sections of A), meaning it cannot be that $y <_2 x$, and hence $I_2 \upharpoonright D_1$ must be an isomorphism onto C', a lower section of B (that is, a lower section of C, but because C is a lower section of B, then C' is also a lower section of B).

Here is why $\mathcal{I}$ in the proof of Theorem 2.9 must be an isomorphism from a lower section of A onto a lower section of B. Recall that $\mathcal{I} = \cup\mathcal{N}$, and $\mathcal{N}$ is the class of all isomorphisms from lower sections of A which are sets onto lower sections of B which are sets. By Proposition 1.4, $\mathcal{I}$ is a function. The domain of $\mathcal{I}$ must be a lower section of A. For suppose it weren't. Then for some x in the domain of $\mathcal{I}$, there would be some $y \in A$ not in the domain of $\mathcal{I}$ such that $x > y$. But x is also in the domain of some $f \in \mathcal{N}$ which is an isomorphism from a lower section of A, which would mean that y is in the domain of f (and thus also of $\cup\mathcal{N}$), a contradiction. Hence the domain of $\mathcal{I}$ is a lower section of A. Similarly, it can be seen that $\mathcal{I}$ is onto a lower section of B.

Section 3

Exercise 3.1

(a): On the one hand, suppose the axiom of substitution. Now, suppose F is a function and x is a set. Then $F \upharpoonright x$ is a subclass of $x \times F''(x)$. But $F''(x)$ is a set by the axiom of substitution, meaning $x \times F''(x)$ is a set, and hence so is $F \upharpoonright x$. On the other hand, suppose that for any function F and any set x, the function $F \upharpoonright x$ is a set. Then consider any function F and any set x. Note that $F''(x)$ is a subclass of $\cup(F \upharpoonright x)$. Thus, because $F \upharpoonright x$ is a set, so is $\cup(F \upharpoonright x)$, and hence so is $F''(x)$.

(b): On the one hand, suppose the axiom of substitution. Now consider some function F whose domain D is a set. Then $F''(D) = \mathrm{Ran}(F)$. Because $F''(D)$ is a set, then, because $D \times F''(D)$ is a set, and F is a subclass of $D \times F''(D)$, F is a set. On the other hand, suppose that any function whose domain is a set is also a set. Now consider any function F and any set x. Then $F''(x)$ is a function whose domain is x, which is a set, and thus $F''(x)$ is also a set by assumption.

Exercise 3.2

Suppose the axiom of choice, and that any class that can be put into a 1-1 correspondence with a set is a set. Now consider any function F and any set x. Let C be the choice function for x. For each $y \in F''(x)$, let $S(y)$ be the set of all $z \in x$ such that $F(z) = y$ (and $S(y)$ is a set, as it is a subset of x, which is a set by assumption). Then $C(S(y)) \in S(y)$, that is, $C(S(y))$ is some $z \in x$ such that $F(z) = y$. Then let f be the set of all ordered pairs $\langle y, C(S(y))\rangle$ for $y \in F''(x)$. This is a 1-1 correspondence between $F''(x)$ and the set of all elements $z \in x$ such that $z = C(S(y))$ for some $y \in F''(x)$. Hence, because $F''(x)$ can be put into a 1-1 correspondence with a set, it is a set.

Section 4

Notes

Note that every well ordered set is also properly well ordered, because obviously every proper lower section of the well ordered set would be a subset of the set, and hence also a set.

Section 5

Notes

Note that, if a class M is well ordered under $\subseteq$, then it is also properly well ordered under $\subseteq$. For consider any proper lower section L of M. Then it is also an initial segment of x for some $x \in M$. Hence, for every $y \in L$, $y \subseteq x$, meaning $L \subseteq \mathcal{P}(x)$, and thus L is a set.

The above explains why M (as defined on page 84) is not only well ordered, but properly well ordered.

Note that, for Theorem 5.1, we of course do not necessarily need that M be minimally superinductive, as long as M is both a set and well orderd under $\subseteq$ such that the three conditions of Theorem 3.3 of Chapter 4 hold.

Here is why, on page 84, if λ is a limit ordinal, then M_λ must be a limit element of M. This follows by the definition of the isomorphism: It cannot be a least element (or else λ would be the least ordinal), and it cannot be a successor element (or else λ would be a successor ordinal). Then, here is why this means that M_λ is the set of all M_α, where $\alpha < \lambda$. For, if $\alpha < \lambda$, then $M_\alpha < M_\lambda$, and hence $M_\alpha \subseteq M_\lambda$. On the other hand, if $x \in M_\lambda$, then $x \in M$ for some $M < M_\lambda$, and it must be that $M = M_\alpha$ for some $\alpha < \lambda$ (or else $M \geq M_\lambda$).

On a side note, it seems to me that, in defining the separation principle, the book really meant to allow $\in$ as the only relation (which would, for instance, be capable of defining $\subseteq$ and $=$ with set quantification). This seems to work, as all sets and classes defined with that principle thus far would only need the relation $\in$ for the definition.

There is a discrepancy in the definitions of an extending operation. The first only says that E must be defined on all ordinal sequences, and that if applied to a sequence of length α, it yields one of length α^+. However, the second definition specifies that E must yield, from a sequence θ_1 of length α, a sequence θ_2 of length α^+ such that $\theta_1 \subseteq \theta_2$ (the second definition is clearly the one being used).

It should be noted that (with the axiom of substitution), every ordinal sequence is a set. Obviously the domain of every ordinal sequence is a set (because the domain is just an ordinal). Thus the range of every ordinal sequence can be put into a 1-1 correspondence with a set, meaning, by the axiom of substitution, the range is also a set. Then, for every ordinal sequence θ, $\theta \subseteq \mathrm{Dom}(\theta) \times \mathrm{Ran}(\theta)$, which is a set. This explains why we can speak of functions whose domain includes all ordinal sequences, as, because all ordinal sequences are sets, we can quantify over them (in the separation princple).

It is remarked that, for every function F on On, $F \restriction \lambda = \cup_{\alpha<\lambda} F \restriction \alpha$ for a limit ordinal λ. Here is why this is. On the one hand, suppose that $x \in F \restriction \lambda$. Then, for some α, $x = \langle \alpha, F(\alpha) \rangle$, and thus $x \in F \restriction \alpha^+$, meaning $x \in \cup_{\alpha<\lambda} F \restriction \alpha$. On the other hand, suppose that $x \in \cup_{\alpha<\lambda} F \restriction \alpha$. Then, for some $\alpha < \lambda$, $x \in F \restriction \alpha$. Consequently $x = \langle \beta, F(\beta) \rangle$ for some $\beta < \alpha$, meaning $\beta < \lambda$, and thus $x \in F \restriction \lambda$ as well. Hence $F \restriction \lambda = \cup_{\alpha<\lambda} F \restriction \alpha$.

Note that the above does not also hold in the case of successor ordinals. That is, $F \restriction \alpha^+ \neq \cup_{\beta<\alpha^+} \restriction \beta$. This is because α is in the domain of $F \restriction \alpha^+$, but for no ordinal β less than α^+ is it the case that α is in the domain of $F \restriction \beta$.

Uniqueness of Functions

Here is why the function in Theorem 5.3 is unique. For suppose we had two functions F and G satisfying the desired conditions. We show by transfinite induction that $F = G$. The case for 0 follows in this way: $F \restriction 1 = E(F \restriction 0) = E(0)$.

Similarly $G \restriction 1 = E(0)$. Thus $F \restriction 1 = G \restriction 1$. But $F \restriction 1 = \langle 0, F(0) \rangle$ and $G \restriction 1 = \langle 0, G(0) \rangle$, and thus $F(0) = G(0)$. Now suppose that, for some α, $F(\beta) = G(\beta)$ for all $\beta < \alpha$. Then $F \restriction \alpha^+ = E(F \restriction \alpha)$ and $G \restriction \alpha^+ = E(G \restriction \alpha)$. But, by assumption, $F \restriction \alpha = G \restriction \alpha$, meaning $F \restriction \alpha^+ = G \restriction \alpha^+$. Hence $\langle \alpha, F(\alpha) \rangle = \langle \alpha, G(\alpha) \rangle$, meaning $F(\alpha) = G(\alpha)$. Thus, by transfinite induction, $F = G$.

Here is why the function in Theorem 5.5 is unqiue. For assume the hypotheses of Theorem 5.5, and that we had two functions F and G such that $F(\alpha) = g(F \restriction \alpha)$ and $G(\alpha) = g(G \restriction \alpha)$ such that the domains of F and G are both On. Then we use transfinite induction to show that $F = G$. The case for the least element holds, as $F(0) = g(F \restriction 0) = g(0) = g(G \restriction 0) = G(0)$. Now suppose that for all ordinals $\beta < \alpha$, $F(\beta) = G(\beta)$. Then $F(\alpha) = g(F \restriction \alpha)$ and $G(\alpha) = g(G \restriction \alpha)$. But, by definition of α, the domain of $F \restriction \alpha$ is every β such that $\beta < \alpha$. Hence, by the assumption that $F(\beta) = G(\beta)$ for every $\beta < \alpha$, $F \restriction \alpha = G \restriction \alpha$, and thus $F(\alpha) = G(\alpha)$. By transfinite induction, it follows that $F(\alpha) = G(\alpha)$ for all α, meaning $F = G$.

Here is the argument showing that the On-sequence in Theorem 5.7 is unique: Suppose we had two On functions F and G satisfying the required conditions. Obviously $F(0) = G(0)$. If $F(\alpha) = G(\alpha)$, then $F(\alpha^+) = g(F(\alpha)) = g(G(\alpha)) = G(\alpha^+)$. If $F(\alpha) = G(\alpha)$ for every $\alpha < \lambda$ for a limit ordinal λ, then $F(\lambda) = \cup_{\alpha<\lambda} F(\alpha) = \cup_{\alpha<\lambda} G(\alpha) = G(\lambda)$. Thus, by transfinite induction, $F = G$.

Lemma 5.2

(1): First, consider some chain C of ordinal sequences. Then $\cup C$ is a function (by Proposition 1.4) and a set. Thus the domain D of $\cup C$ is a set. Now suppose $\alpha \in D$ and $\beta \in \alpha$. Then, for some $f \in C$, $\alpha \in C$. But $\beta < \alpha$, and by definition of f, it follows that $\beta \in f$ (and thus $\beta \in D$). Hence D is transitive, meaning, because it is a transitive set of ordinals, it is also some ordinal α. Then $\cup C$ is an ordinal sequence of length α.

Suppose that λ is a limit ordinal such that all elements of C are of length less than λ, and if for every $\alpha < \lambda$ there is a an element of C of length α. We need to show that $\cup C$ is of length λ, that is, $\alpha \in \mathrm{Dom}(\cup C)$ iff $\alpha < \lambda$. On the one hand, suppose that $\alpha \in \mathrm{Dom}(\cup C)$. Then $\alpha \in f$ for some $f \in C$. By definition of C, f is of a length less than λ, meaning $\alpha < \lambda$. On the other hand, suppose that $\alpha < \lambda$. Then there is an element of C length α. Then there also must be an element f of C of length α^+, as $\alpha^+ < \lambda$ (or else λ would not be a limit element), and $\alpha \in f$, that is, $\alpha \in \mathrm{Dom}(\cup C)$. Thus $\cup C$ is of length λ.

(2): Suppose that $\mathcal{N}$ is a nest of ordinal sequences and that $\mathcal{N}$ is not a set. Again, by Proposition 1.4, $\cup \mathcal{N}$ is a function. First, we show that $\mathcal{N}$ is properly well ordered under $\subseteq$. Because it is a nest, it is linearly ordered under $\subseteq$. Now consider any subclass C of $\mathcal{N}$. Then consider the class F of ordinals α such that a function of length α is in C. Then F has a least element β. Then the function f which is of length β is clearly the least element of C. Hence $\mathcal{N}$ is well ordered under $\subseteq$ (because the well ordering is under $\subseteq$, it is also properly well ordered under $\subseteq$). Then, because $\mathcal{N}$ is not a set, by the counting theorem, there is an isomorphism from $\mathcal{N}$ onto On. Now consider the class A of all ordinals in the domain of $\cup \mathcal{N}$. An obvious transfinite induction argument shows that $A = On$. This obviously implies that the domain of $\cup \mathcal{N}$ is On.

Theorem 5.3

Here is why we know that there is a minimally superinductive class under E (which is an extending operation, whose domain includes all ordinal sequences). In order to use the necessary theorem presented in chapter 4, we would need to know that (1) E is defined on 0; (2) if E is defined on an ordinal sequence x, then E is defined on $E(x)$; (3) for any chain C of ordinal sequences that E is defined on, E is defined on $\cup C$; (4) E is progressing for all ordinal sequences. (1) follows because 0 is an ordinal sequence. (2) follows because E is an extending operation. (3) Consider some chain C of ordinal sequences that E is defined on. Then, by Lemma 5.2 (1), $\cup C$ is an ordinal sequence, and thus E is defined for $\cup C$. (4) follows by the definition of an extending operation.

Note that this above argument acutally shows that there is a class that is minimally inductive under $E \restriction On$, but this class is of course also minimally inductive under E.

Here is why we can conclude that, if λ is a limit ordinal and if for each $\alpha < \lambda$, M_α is an α-sequence, then M_λ is a λ sequence. First, note that, by Theorem 5.1, $M_\lambda = \cup_{\alpha<\lambda} M_\alpha$. Thus M_λ is the union of a chain C of ordinal sequences. Clearly all elements of C are of length less than λ (by the induction assumption). It remains to show (in order to use part (1) of Lemma 5.2) that for every $\alpha < \lambda$ there is an element of C of length α. This can be proven by transfinite induction on

the ordinals: we show that if $\alpha < \lambda$, then there is an ordinal sequence of length α in C. This holds for 0, as by definition of M, $0 \in M$ (and thus also in C). For any α, if $\alpha < \lambda$ and there is some ordinal sequence of length α in M, then, if $\alpha^+ < \lambda$, by definition of M and the induction assumption, there is an ordinal sequence of length α^+ in M (namely M_{α^+}). If for all $\alpha < \lambda_1$ for some limit ordinal λ_1, there is some ordinal sequence of length α in M, then, if $\lambda_1 < \lambda$, by definition of M and the induction assumption, there is an ordinal sequence (namely M_{λ_1}) of length λ_1 in M. Thus, for every $\alpha < \lambda$, there is an element of C of length α, meaning $\cup C$, that is M_λ, is of length λ.

Theorem 5.8

The proof of this theorem is incorrect (and the theorem itself is incorrect). Now, the proof says: "By Theorem 5.5, taking $g(x) = h(\text{Ran}(x))$, if x is a function (otherwise take $g(x) = x$)." However, this is not correct, as it very well may be that h is not defined for $\text{Ran}(x)$ for some function x. A counter example to this theorem is simply the function $f = \{\langle 1, 1\rangle\}$. For, by the Theorem, we would have a function F such that $F(0) = f(F''(0)) = f(0)$, but 0 is not defined for f.

If we added that h must be defined on all sets as a prerequisite to the theorem, then the theorem obviously is correct.

Exercise 5.1

I could not prove this using Theorem 5.8 (or Theorem 5.7). The following proves that, if part (1) of the counting theorem holds, then we can use Theorem 5.7 to prove part (2) (and Theorem 5.8 basically allows us to make the same argument).

(1): For the first part of the theorem, we wish to show that every well ordered set is isomorphic to exactly one ordinal number α. By the comparability theorem, either A is isomorphic to a lower section of On, or On is isomorphic to a lower section of A. Suppose that On was isomorphic to a lower section L of A under ϕ. We define g as follows: if x is a proper lower section of A, and y is the least element of $A - x$, then let $g(x) = x \cup \{y\}$. Otherwise take $g(x) = x$. By Theorem 5.7, there is an On-sequence $S_0, S_1, ..., S_\alpha, ...$, such that:
(1) $S_0 = \emptyset$;
(2) for each α, $S_{\alpha+1} = g(S_\alpha)$;
(3) for each limit ordinal λ, $S_\lambda = \cup_{\alpha<\lambda} S_\alpha$.
Let S be the class of all S_α's. It is a set as it is a subset of $\mathcal{P}(A)$. We aim to show that S is minimally superinductive under g, and hence not a set, which is a contradiction. By (1), $\emptyset \in S$. Also, by (2), S is closed under g. Suppose now we have some chain C of elements of S. Then if C contains a greatest element, clearly $\cup C \in S$, as $\cup C$ is just the greatest element of C. On the other hand, if C contains no greatest element, I was unable to show that $\cup C \in S$.

(2): For the second part of the theorem, we wish to show that every properly well ordered proper class is isomorphic to the class On of all ordinals. By the comparability theorem, either A is isomorphic to a lower section of On, or On is isomorphic to a lower section of A. Suppose that A was isomorphic to some lower section of On. Then some ordinal γ would be isomorphic to A under some function ϕ. Define the function g as follows: if x is a proper lower section of A, and y is the least element of $A - x$, then let $g(x) = x \cup \{y\}$. Otherwise take $g(x) = x$. By Theorem 5.7, there is an On-sequence $S_0, S_1, ..., S_\alpha, ...$, such that:
(1) $S_0 = \emptyset$;
(2) for each α, $S_{\alpha+1} = g(S_\alpha)$;
(3) for each limit ordinal λ, $S_\lambda = \cup_{\alpha<\lambda} S_\alpha$.
We wish to show that $S_\gamma = A$, and thus A would be a set, a contradiction. Note that the On-sequence above is obviously such that every S_α is a lower section (perhaps proper) of A (this can be seen by using transfinite induction). Also, it can be seen that if $\alpha < \gamma$, then S_α is a proper lower section of A (one way to directly show this is first by showing that if $\alpha < \gamma$, then S_α can be placed in a 1-1 correspondence with α).

Lemma 0.1. For every ordinal α such that $\alpha < \gamma$, $S_{\alpha^+} = S_\alpha \cup \phi(\alpha)$.

Proof. We prove this by transfinite induction. Clearly this holds for $\emptyset$. Suppose it holds for some α. Thus $S_{\alpha^+} = S_\alpha \cup \phi(\alpha)$. Suppose that $\alpha^+ < \gamma$. But then $\phi(\alpha^+)$ is the least element outside of $A - S_{\alpha^+}$, meaning $S_{\alpha^{++}} = S_{\alpha^+} \cup \phi(\alpha^+)$. On the other hand, suppose it holds for all $\alpha < \lambda$ for a limit ordinal λ such that $\lambda < \gamma$. Then $S_\lambda = \cup_{\alpha<\lambda} S_\alpha$, meaning for every $\alpha < \lambda$, $\phi(\alpha) \in S_\lambda$. Hence the least element of $A - S_\lambda$ is $\phi(\lambda)$, meaning $S_{\lambda^+} = S_\lambda \cup \{\phi(\lambda)\}$. □

We first show that $x \in S_\alpha$ iff the β for which $x = \phi(\beta)$ is such that $\beta < \alpha$.

Proof. (a) On the one hand, suppose that $x = \phi(\beta)$ for some β. We show that if $\beta < \alpha$, then $x \in S_\alpha$. β cannot be less than 0, meaning the case for S_0 vacuously holds. Now suppose that some successor ordinal α^+ is such that $\beta < \alpha^+$, and that the result holds for α. Then, using Lemma 0.1, $x \in S_{\alpha^+}$. Now suppose that, for some limit ordinal λ, the result holds for all ordinals $\alpha < \lambda$ and that $\beta < \lambda$. Then $\beta^+ < \lambda$, and by assumption, $x \in S_{\beta^+}$. Because $S_\lambda = \cup_{\alpha<\lambda} S_\alpha$, it then follows that $x \in S_\lambda$. Thus, by transfinite induction, $x \in S_\alpha$ for all $\alpha > \beta$.

(b) On the other hand, we show that if $x \in S_\alpha$, then the β for which $x = \phi(\beta)$ is such that $\beta < \alpha$. We show by transfinite induction that this holds for all S_α. It vacuously holds for S_0. Suppose it holds for some α. Using this assumption, as well as part (a) of this proof, we have that $x \in S_\alpha$ iff the β for which $x = \phi(\beta)$ is such that $\beta < \alpha$. Consider the β for which the least element of $A - S_\alpha$ (if there is one) is equal to $\phi(\beta)$. Then β contains only ordinals less than α. For suppose that β contained some ordinal $\delta \geq \alpha$. Now, $\phi(\delta) \notin S_\alpha$ (by assumption), meaning $\phi(\delta) \in A - S_\alpha$. But $\delta < \beta$ and thus $\phi(\delta)$ preceedes $\phi(\beta)$, contrary to the assumption that $\phi(\beta)$ was the least element of $A - S_\alpha$. Thus β contains only ordinals less than α, meaning $\beta < \alpha^+$. Now suppose that, for some limit ordinal λ, the result holds for all S_α for $\alpha < \lambda$, and that $\beta < \lambda$. Then if $x \in S_\lambda$, $x \in S_\alpha$ for some $\alpha < \lambda$, meaning the β for which $x = \phi(\beta)$ is such that $\beta < \alpha < \lambda$ (by assumption). □

Thus $x \in S_\alpha$ iff the β for which $x = \phi(\beta)$ is such that $\beta < \alpha$. If $x \in A$, then the β for whcih $x = \phi(\beta)$ is such that $\beta < \gamma$, meaning $x \in S_\gamma$. On the other hand, if $x \in S_\gamma$, then obviously $x \in A$ (because S_γ is a lower section of A). Thus $S_\gamma = A$. But this would mean A was a set, contrary to the assumption that it wasn't. Hence it cannot be that A is isomorphic to a lower section of On. Thus On must be isomorphic to a lower section of A. But it cannot be a lower section of A; for, because A is properly well ordered, On would then be isomorphic to a set, which cannot occur (by part (1) of this exercise), meaning On must be isomorphic to the whole of A.

Exercise 5.2

The problem, as stated, is clearly incorrect. For we can obviously find functions that are not minimally superinductive under any class. For the problem, I assume that g is a progressing function and that (1) g is defined on $\emptyset$; (2) if g is defined on x then g is defined on $g(x)$; (3) for any chain C of elements that g is defined for, g must be defined on $\cup C$.

Consider any function g satisfying the above three conditions. By Theorem 5.7, there is an On-sequence $S_0, S_1, ..., S_\alpha, ...$ such that:
(1) $S_0 = \emptyset$;
(2) for each α, $S_{\alpha+1} = g(S_\alpha)$;
(3) for each limit ordinal λ, $S_\lambda = \cup_{\alpha<\lambda} S_\alpha$.
We aim to show that the class S of all S_α for all ordinals α is minimally superinductive under g.

(1') By (1), $0 \in S$.

(2') Suppose that $x \in S$. Then, for some α, $x = S_\alpha$. Then $g(x) \in S$ by (2).

Lemma 0.2. For any non-empty chain C of ordinals containing no greatest element, $\cup C$ is a limit ordinal

Proof. Consider some chain C of ordinals containing no greatest element. Obviously $\cup C \neq \emptyset$. Suppose that $\cup C$ was a successor ordinal for some α. Then $\cup C = \alpha \cup \{\alpha\}$, in which case α would be the greatest element of $\cup C$. Then consider the least ordinal β such that $\beta \in C$ and $\alpha \in \beta$. Clearly β must be the greatest element of C, as if there was an element in C greater than β, then β would be in $\cup C$, contrary to the assumption that α was the greatest element of $\cup C$. Hence C contains a greatest element, a contradiction. Thus $\cup C$ is neither the least ordinal nor a successor ordinal, and is consequently a limit ordinal. □

(3') Consider some chain C of elements of S. Consider the class O of ordinals α such that $S_\alpha \in C$. O is a set, as it can be placed in a 1-1 correspondence with C. If O contains a greatest ordinal α, then $\cup C = S_\alpha$. If it does not contain a greatest ordinal, then note that $\cup O$ is a limit ordinal by the preceeding Lemma. Let $\lambda = \cup O$. Hence, by (3), $S_\lambda = \cup_{\alpha<\lambda} S_\alpha$. We wish to show that $S_\lambda = \cup C$. Obviously, if $x \in \cup C$, then $x \in S_\lambda$. On the other hand, suppose that $x \in S_\lambda$. Thus, for

some $\alpha < \lambda$, $x \in S_\alpha$. We know that $\alpha \in \lambda$, that is, $\alpha \in \cup O$. Hence, for some $\beta \in O$, $\alpha \in \beta$. But $S_\alpha \subseteq S_\beta$ if $\alpha < \beta$ (this can easily be seen, assuming g is progressing; in fact, this is just Exercise 5.4), meaning $x \in S_\beta$ and hence $x \in \cup C$. Thus $\cup C = S_\lambda$, meaning $\cup C \in S$.

Hence we have shown that S is superinductive under g. It remains to be shown that S is minimally superinductive under g. Consider any class C that is superinductive under g. We can show by transfinite induction on the On-sequence $S_0, S_1, ..., S_\alpha, ...$ that, for every α, $S_\alpha \in C$, meaning $S \subseteq C$. Obviously $S_0 \in C$. Suppose that some $S_\alpha \in C$. Then $g(S_\alpha) \in C$, that is, $S_{\alpha^+} \in C$. Now suppose that for some limit ordinal λ, for all $\alpha < \lambda$, $S_\alpha \in C$. Then $S_\lambda = \cup_{\alpha<\lambda} S_\alpha$, that is, S_λ is a chain of elements of C, meaning $S_\lambda \in C$. Hence, for all ordinals α, $S_\alpha \in C$, meaning $S \subseteq C$. Thus we have shown that S is minimally inductive under g, therefore completing the proof.

Exercise 5.3

Note that, for this problem, I will assume that, in order to use Theorem 5.8 on some function h, h must be defined for all sets (as is done in this proof).

(a): Suppose, on the contrary, that for all ordinals α, $F(\alpha) \in S$. We show that this would mean On was a set.

We now show that, if $\alpha \neq \beta$, then $F(\alpha) \neq F(\beta)$. Without loss of generality, suppose that $\alpha < \beta$. Then $F(\alpha) \in F''(\beta)$. But $F(\beta) = h(F''(\beta))$. By assumption, $F''(\beta)$ must be a proper subset of S, so that $F(\beta) = h(F''(\beta)) = C(S - F''(\beta))$, meaning it cannot be that $F(\beta) = F(\alpha)$ (because $F(\alpha) \notin S - F''(\beta)$) (Note that this argument shows that, if $\alpha \neq \beta$ and $F(\alpha) \in S$ and $F(\beta) \in S$, then $F(\alpha) \neq F(\beta)$. This fact will be used in part (b)).

Thus, for all α and β, $F(\alpha) \neq F(\beta)$. Hence the class F of $F(\alpha)$ for all α can be put into a 1-1 correspondence with On. But, because $F(\alpha) \in S$ for all α, F is a set. Thus, by the axiom of substitution, On would be a set, which is impossible.

Thus it cannot be that $F(\alpha) \in S$ for all $F(\alpha)$, meaning $F(\delta) \notin S$ for some δ.

(b): Let β be the least ordinal such that $F(\beta) \notin S$. Because $F(\beta) = h(F''(\beta))$, either $F''(\beta) = S$, or $F''(\beta)$ is not a subset of S. Now, clearly $F''(\beta) \subseteq S$. For, by the assumption that β is the least ordinal such that $F(\beta) \notin S$, for all $\alpha < \beta$, $F(\alpha) \in S$. But $F''(\beta) = \{F_\alpha | \alpha < \beta\}$, meaning $F''(\beta) \subseteq S$. Thus, because it is the case that $F''(\beta) \subseteq S$, and either $F''(\beta) = S$ or $F''(\beta)$ is not a subset of S, it follows that $F''(\beta) = S$. Next, we show that $F \restriction \beta$ is 1-1. For all $\alpha_1 \in \beta$ and $\alpha_2 \in \beta$, if $\alpha_1 \neq \alpha_2$, then $F(\alpha_1) \neq F(\alpha_2)$ (as shown in part (a)). This shows that $F \restriction \beta$ is 1-1, meaning β can be put into a 1-1 correspondence with S, and so S can be well ordered.

This completes the proof that every set which has a choice function can be well ordered.

Exercise 5.4

We aim to show that for any ordinals α and β, if $\alpha \leq \beta$ then $F(\alpha) \subseteq F(\beta)$. This can be proven by transfinite induction on the ordinals. That is, we show that, for all β, $F(\alpha) \subseteq F(\beta)$ holds for all $\alpha \leq \beta$. The case for 0 obviously holds. On the one hand, suppose that the result holds for some α. Then by (a), $F(\alpha) \subseteq F(\alpha^+)$. By assumption, for all $\beta \leq \alpha$, $F(\beta) \subseteq F(\alpha) \subseteq F(\alpha^+)$. Hence for all $\beta \leq \alpha^+$, $F(\beta) \subseteq F(\alpha^+)$. On the other hand, for any limit ordinal λ, by (b), for all $\alpha < \lambda$, $F(\alpha) \subseteq F(\lambda)$.

Section 6

Notes

On page 88, here is why the well orderind described is really a well ordering. In particular, we are attempting to evaluate $\alpha + \beta$ where α and β are ordinals. We take two well ordered sets a and b such that a is isomorphic to α, b is isomorphic to β, and a and b are disjoint. Let $\leq_1$ and $\leq_2$ be the well orderings on a and b, respectively. Then we define the

ordering $\prec$ as follows: if $x, y \in a, x \prec y$ iff $x < y$ in the ordering on a; if $x, y \in b, x \prec y$ iff $x < y$ in the ordering on b; if $x \in a$ and $y \in b$, $x \prec y$ is true; and if $x \in b$ and $y \in a$, $x \prec y$ is false. We show that $\prec$ is a well ordering for $a \cup b$.

(1) Obviously $\prec$ is reflexive. For if $x \in a$, then $x \leq_1 x$ and thus $x \prec x$, and if $x \in b$, then $x \leq_2 x$ and thus $x \prec x$.

(2) The relation is also anti-symmetric. For if both $x \prec y$ and $y \prec x$, then either both of x, y are in a, or both of x, y are in b, meaning $x = y$ by the assumption of $\leq_1$ and $\leq_2$ being well orderings.

(3) The relation is also transitive. For if $x \prec y$ and $y \prec z$, then either (1) all of x, y, z are in a; (2) all of x, y, z are in b; (3) x, y are in a and z is in b; (4) x is in a and y, z are in b. In case (1), $x \prec z$ by the well ordering of a. Similarly case (2) implies $x \prec z$. Case (3) implies $x \prec z$ because $u \prec v$ holds if $u \in a$ and $z \in b$. Case (4) implies $x \prec z$ holds for the same reason as case (3). Thus $\prec$ is transitive.

(4) The relation is also linear. Obviously this is the case if x, y are either both in a or b. On the other hand, suppose that x, y are not both in a and not both in b. Without loss of generality, suppose $x \in a$. Then $x \prec y$. Thus $\prec$ is a linear ordering on $a \cup b$.

(5) Lastly, we must show that it is a well ordering on $a \cup b$. Consider any subclass C of $a \cup b$. If C contains any elements of a, then the class of all elements of a in C contains a least element (under $\leq_1$) e. Then e is the least element of C. If C does not contain any elements of a, then C is a subclass of b, hence contains a least element (under $\leq_2$), which is also the least element of C under $\prec$. Thus $\prec$ is a well ordering on $a \cup b$.

It should be noted that obvious transfinite induction arguments show that ordinal addition, multiplication, and exponentiation are all closed under On.

Theorem 6.1

The book claims that the proof of Theorem 6.1 is proved in "essentially the same way" as Theorem 5.6 - this I could not verify. Instead, I was only able to prove this in a similar manner to the double analogue of Theorem 8.1 in Chapter 3.

Theorem 0.3. For any functions h (whose domain includes $(On \times On) \times On$), f and g (whose domains include On), there is a unique two-place function F defined on all ordinals, such that:
(1) $F(\alpha, 0) = g(\alpha)$;
(2) $F(\alpha, \beta^+) = h(F(\alpha, \beta), \alpha, \beta)$;
(3) for a limit ordinal λ, $F(\alpha, \lambda) = f(\{F(\alpha, \beta) | \beta < \lambda\})$.

Proof. This can be proven in the following way. We first show that for every ordinal α, there is a function $f_\alpha(\beta, \gamma)$ defined on all ordinals β and all ordinals $\gamma \leq \alpha$ such that $(1')$ $f_\alpha(\beta, 0) = g(\alpha)$ (for all β); $(2')$ for all ordinals β and for each ordinal $\gamma < \alpha$, $f_\alpha(\beta, \gamma^+) = h(f_\alpha(\beta, \gamma), \beta, \gamma)$; $(3')$ for every limit ordinal $\lambda \leq \alpha$, $f_\alpha(\beta, \lambda) = f(\{f_\lambda(\gamma, \beta) | \gamma < \lambda\})$. We prove this by transfinite induction. The base case is when $\alpha = 0$. Then we just take f_0 to be the class of all triples $\langle \beta, 0, g(\beta) \rangle$ (for all ordinals β). Now suppose the result holds for some α. Then we take f_{α^+} to be the class of all triples that f_α was defined on, as well as the triples $\langle \beta, \alpha^+, h(f_\alpha(\beta, \alpha), \alpha, \beta) \rangle$ for all ordinals β. On the other hand, suppose that the result holds for all ordinals less than some limit ordinal λ. Then take f_λ to be the set of all triples $\langle \alpha, \beta, f_\beta(\alpha, \beta) \rangle$ for all ordinals α and β, where $\beta < \lambda$, as well as all triples $\langle \alpha, \lambda, f(\{f_\beta(\alpha, \beta)\} | \beta < \lambda\})$ for all ordinals α. An obvious transfinite induction argument shows that this definition of f_α really works.

Next we show that for each α, $f_\alpha \subseteq f_{\alpha^+}$ and that for each limit ordinal λ and all $\beta < \lambda$, $f_\beta \subseteq f_\lambda$. First, we show that $f_\alpha \subseteq f_{\alpha^+}$, which is to say, for every ordinal $\beta \leq \alpha$, $f_\alpha(\gamma, \beta) = f_{\alpha^+}(\gamma, \beta)$ holds for all ordinals γ. This is shown by an obvious transfinite induction argument. Similarly, it can easily be shown that for each limit ordinal λ and all $\beta < \lambda$, $f_\beta \subseteq f_\lambda$. Thus $f_\alpha(\gamma, \beta) = f_{\alpha^+}(\gamma, \beta)$ for all $\beta \leq \alpha$ and all γ, and $f_\alpha(\gamma, \beta) = f_\lambda(\gamma, \beta)$, for a limit ordinal λ, holds for all $\alpha < \lambda$, all γ, and all $\beta \leq \alpha$.

Then take F to be the class of all triples $\langle \alpha, \beta, f_\beta(\beta, \alpha) \rangle$ (for all α and β). We need to show that the three conditions of the theorem hold. (1) holds, as $F(\alpha, 0) = f_0(\alpha, 0) = g(\alpha)$. (2) holds, as

$$F(\alpha, \beta^+) = f_{\beta^+}(\alpha, \beta^+) = h(f_{\beta^+}(\alpha, \beta), \alpha, \beta) = h(f_\beta(\alpha, \beta), \alpha, \beta) = h(F(\alpha, \beta), \alpha, \beta)$$

(3) holds, as

$$F(\alpha, \lambda) = f(\{f_\lambda(\gamma, \beta) | \beta < \lambda\}) = f(\{f_\beta(\gamma, \beta) | \beta < \lambda\}) = f(\{F(\gamma, \beta) | \beta < \lambda\})$$

Thus F is the desired function. $\square$

Remark: When we are defining f_λ in the limit ordinal case, note that $\{f_\beta(\alpha,\beta)|\beta<\lambda\}$ is a set because it can be placed in a 1-1 correspondence with λ. This is why $f(\{f_\beta(\alpha,\beta)\}|\beta<\lambda\})$ is defined. Also, an easy transfinite induction argument shows that F is a unique function. The original Theorem 5.6 can also be proven using a similar idea as the above; but, as was done in the book, if the various other forms of the trasfinite recursion theorem are proven first, the proof of Theorem 5.6 follows more easily.

Lemmas

Lemma 0.4. If $\beta<\gamma$, then $\alpha\times\beta<\alpha\times\gamma, \alpha+\beta<\alpha+\gamma$, and $\alpha^\beta<\alpha^\gamma$. The cases for multiplication and exponentiation require $\alpha\neq 0$. The case for exponentiation also requires $\alpha\neq 1$.

Proof. We give the proof using transfinite induction for the addition case. The proofs for multiplciation and exponentiation are similar.

We prove this by induction on γ. This vacuously holds for 0. Now suppose it holds for some γ. Then $\alpha+\gamma^+ = \alpha+\gamma\cup\{\alpha+\gamma\}$. By the induction assumption, for all $\beta<\gamma$, $\alpha+\beta<\alpha+\gamma<\alpha+\gamma^+$. The only other $\beta<\gamma^+$ is $\alpha+\gamma$ itself, which is less than $\alpha+\gamma^+$ by definition of $\alpha+\gamma^+$.

On the other hand, suppose that it holds for all ordinals less than some limit ordinal λ. Then $\alpha+\lambda=\cup_{\gamma<\lambda}(\alpha+\gamma)$. Then suppose $\beta<\lambda$. Now, $\alpha+\beta<\alpha+\beta^+$. But also $\alpha+\beta^+<\alpha+\lambda$, meaning $\alpha+\beta<\alpha+\lambda$. $\square$

Lemma 0.5. If λ is a limit ordinal, then $\alpha\times\lambda, \alpha+\lambda$ and α^λ are all limit ordinals (in the cases of multiplication and exponentiation, for all ordinals $\alpha>0$, and, in the case of exponentiation, for all ordinals $\alpha>1$).

Proof. We give the proof for addition. The proofs for multiplication and exponentiation are similar.

Suppose that λ is a limit ordinal. Clearly $\alpha+\lambda$ is not the least ordinal. Suppose that $\alpha+\lambda$ was a successor ordinal of some δ. Then $\alpha+\lambda=\cup_{\gamma<\lambda}(\alpha+\gamma)=\delta\cup\{\delta\}$. Hence, for some $\gamma<\lambda$, $\delta<\alpha+\gamma$. Then $\delta^+\leq\alpha+\gamma$. But then $\delta^+<\alpha+\gamma^+$, and thus $\delta^+\in\alpha+\lambda$, contradicting the fact that δ is the greatest element of $\alpha+\lambda$. Hence $\alpha+\lambda$ cannot be a successor ordinal, and is consequently a limit ordinal. $\square$

Exercise 6.1

(a) We prove this by transfinite induction on α. The case for 0 holds as $0+0=0$ by definition. Suppose it holds for some α. Then $0+\alpha^+=(0+\alpha)^+=\alpha^+$. Now suppose that, for some limit ordinal λ, it holds for all $\alpha<\lambda$. Then $0+\lambda=\cup_{\alpha<\lambda}(0+\alpha)=\cup_{\alpha<\lambda}\alpha=\lambda$. Hence the proof is complete.

(b) We prove this by transfinite induction on β. That is, we show that, for all ordinals β, $\alpha\leq\alpha+\beta$ holds. The case for 0 holds, as $\alpha+0=\alpha$. Suppose that the result holds for some β. Then $\alpha\leq\alpha+\beta$. Also $\alpha+\beta^+=(\alpha+\beta)^+$, and thus $\alpha\leq\alpha+\beta^+$. Now suppose that for all $\beta<\lambda$ for a limit ordinal λ, the result holds. Thus $\alpha\leq\alpha+\beta$ for all $\beta<\lambda$. Then $\alpha+\lambda=\cup_{\beta<\lambda}(\alpha+\beta)$, and clearly this means $\alpha\leq\alpha+\lambda$.

(c) We prove this by transfinite induction on γ. The case for 0 holds, as $\alpha+(\beta+0)=\alpha+\beta$ and $(\alpha+\beta)+0=\alpha+\beta$. Suppose the result holds for some γ. Then

$$\alpha+(\beta+\gamma^+)=\alpha+(\beta+\gamma)^+=(\alpha+(\beta+\gamma))^+=((\alpha+\beta)+\gamma)^+$$

and

$$(\alpha+\beta)+\gamma^+=((\alpha+\beta)+\gamma)^+$$

that is, $\alpha+(\beta+\gamma^+)=(\alpha+\beta)+\gamma^+$. Now suppose the result holds for all $\gamma<\lambda$ for some limit ordinal λ. Then

$$\alpha+(\beta+\lambda)=\alpha+\cup_{\gamma<\lambda}(\beta+\gamma)$$

and

$$(\alpha + \beta) + \lambda = \cup_{\gamma<\lambda}((\alpha + \beta) + \gamma)$$
$$= \cup_{\gamma<\lambda}(\alpha + (\beta + \gamma)) \text{ (by the induction assumption)}$$

Let $O = \cup_{\gamma<\lambda}(\beta + \gamma)$. By Lemma 0.5, O is a limit ordinal. We now aim to show that $\alpha + \cup_{\gamma<\lambda}(\beta + \gamma) = \cup_{\gamma<\lambda}(\alpha + (\beta + \gamma))$. Note that, because O is a limit ordinal, $\alpha + \cup_{\gamma<\lambda}(\beta + \gamma) = \cup_{\delta<O}(\alpha + \delta)$.

On the one hand, suppose that $x \in \cup_{\delta<O}(\alpha + \delta)$. Then $x \in \alpha + \delta$ for some $\delta \in O$, that is, for some $\delta \in \cup_{\gamma<\lambda}(\beta + \gamma)$. Thus, for some $\beta + \gamma$ where $\gamma < \lambda$, $\delta < \beta + \gamma$, meaning $\alpha + \delta < \alpha + (\beta + \gamma)$ (by Lemma 0.4) thus $\alpha + \delta \in \cup_{\gamma<\lambda}(\alpha + (\beta + \gamma))$.

On the other hand, suppose that $x \in \cup_{\gamma<\lambda}(\alpha + (\beta + \gamma))$. Then, for some $\gamma < \lambda$, $x \in \alpha + (\beta + \gamma)$. But $\beta + \gamma < O$, meaning $x \in \cup_{\delta<O}(\alpha + \delta)$. Thus $\cup_{\delta<O}(\alpha + \delta) = \cup_{\gamma<\lambda}(\alpha + (\beta + \gamma))$, therefore completing the transfinite induction.

(d) Suppose $\alpha \leq \beta$. We prove this by induction on γ. The case for 0 holds, as $\alpha + 0 = \alpha$ and $\beta + 0 = \beta$, meaning $\alpha + 0 \leq \beta + 0$. Suppose it holds for some γ. Then $\alpha + \gamma^+ = (\alpha + \gamma)^+$ and $\beta + \gamma^+ = (\beta + \gamma)^+$, meaning $\alpha + \gamma^+ \leq \beta + \gamma^+$. Now suppose the result holds for all $\gamma < \lambda$ for a limit ordinal λ. Then $\alpha + \lambda = \cup_{\gamma<\lambda}(\alpha + \gamma)$ and $\beta + \lambda = \cup_{\gamma<\lambda}(\beta + \gamma)$, meaning $\alpha + \lambda \leq \beta + \lambda$. $\leq$ cannot be replaced by $<$. For instance, $1 < 2$, but $1 + \omega = \omega$ and $2 + \omega = \omega$, meaning only $1 + \omega \leq 2 + \omega$.

Exercise 6.2

(a)

$$\omega \times 2 = (\omega \times 1) + \omega = \omega + \omega = \cup_{\alpha<\omega}(\omega + \alpha)$$

but

$$2 \times \omega = \cup_{\alpha<\omega}(2 \times \alpha)$$

From the above, $\omega \in \omega \times 2$, but $\omega \notin 2 \times \omega$, meaning $\omega \times 2 \neq 2 \times \omega$.

(b)

$$\alpha \times 1 = (\alpha \times 0) + \alpha = 0 + \alpha = \alpha$$

We prove by transfinite induction on α that $1 \times \alpha = \alpha$. Clearly this holds for 0. Suppose it holds for some α. Then $1 \times \alpha^+ = (1 \times \alpha) + 1 = \alpha + 1 = \alpha^+$. Now suppose it holds for all $\alpha < \lambda$ for a limit ordinal λ. Then $1 \times \lambda = \cup_{\alpha<\lambda}(1 \times \alpha) = \cup_{\alpha<\lambda}\alpha = \lambda$. Thus $1 \times \alpha = \alpha$ holds for all α, meaning $1 \times \alpha = \alpha \times 1 = \alpha$.

Lemma 0.6. Ordinal multiplication is left-distributive.

Proof. We show that ordinal multiplication is left-distributive. That is, we show that $\alpha \times (\beta + \gamma) = \alpha \times \beta + \alpha \times \gamma$. We prove this by transfinite induction on γ. Clearly this holds for 0 and 1. Now suppose it holds for all ordinals less than $\delta > 1$. On the one hand, suppose that δ is a successor ordinal of some γ. Then

$$\alpha \times (\beta + \gamma^+) = \alpha \times (\beta + \gamma)^+$$
$$= (\alpha \times (\beta + \gamma)) + \alpha$$
$$= (\alpha \times \beta + \alpha \times \gamma) + \alpha$$
$$= \alpha \times \beta + (\alpha \times \gamma + \alpha) \text{ (Associativity of ordinal addition)}$$
$$= \alpha \times \beta + \alpha \times \gamma^+ \text{ (Transfinite induction assumption)}$$

On the other hand, suppose that δ is a limit ordinal λ. Then

$$\alpha \times (\beta + \lambda) = \alpha \times \cup_{\gamma<\lambda}(\beta + \gamma)$$

Let $O = \cup_{\gamma<\lambda}(\beta + \gamma)$. O is a limit ordinal by Lemma 0.5. Thus

$$\alpha \times O = \cup_{\delta<O}(\alpha \times \delta)$$

Also

$$\alpha \times \beta + \alpha \times \lambda = \alpha \times \beta + \cup_{\gamma<\lambda}(\alpha \times \gamma)$$

Let $J = \cup_{\gamma<\lambda}(\alpha \times \gamma)$, which is also a limit ordinal by Lemma 0.5. Thus

$$\alpha \times \beta + \cup_{\gamma<\lambda}(\alpha \times \gamma) = \cup_{\epsilon<J}(\alpha \times \beta + \epsilon)$$

We need to show that $\alpha \times O = \alpha \times \beta + J$, that is,

$$\cup_{\delta<O}(\alpha \times \delta) = \cup_{\epsilon<J}(\alpha \times \beta + \epsilon)$$

On the one hand, suppose that $x \in \cup_{\delta<O}(\alpha \times \delta)$. Then, for some $\delta < O$, $x < \alpha \times \delta$. By definition of O, $\delta < \beta + \gamma$ for some $\gamma < \lambda$. Thus $x < \alpha \times \delta \leq \alpha \times (\beta + \gamma) = \alpha \times \beta + \alpha \times \gamma$ by the induction assumption. Now, $\alpha \times \beta + \alpha \times \gamma \subseteq \cup_{\epsilon<J}(\alpha \times \beta + \epsilon)$. Hence $x \in \cup_{\epsilon<J}(\alpha \times \beta + \epsilon)$, therefore showing that $\cup_{\delta<O}(\alpha \times \delta) \subseteq \cup_{\epsilon<J}(\alpha \times \beta + \epsilon)$.

On the other hand, suppose that $x \in \cup_{\epsilon<J}(\alpha \times \beta + \epsilon)$. Then $x < \alpha \times \beta + \epsilon$ for some $\epsilon < J$. By definition of J, $\epsilon < \alpha \times \gamma$ for some $\gamma < \lambda$. Thus $x < \alpha \times \beta + \epsilon < \alpha \times \beta + \alpha \times \gamma = \alpha \times (\beta + \gamma)$ by the induction assumption. Now, $\alpha \times (\beta + \gamma) \subseteq \cup_{\delta<O}(\alpha \times \delta)$. Hence $x \in \cup_{\delta<O}(\alpha \times \delta)$, therefore showing that $\cup_{\epsilon<J}(\alpha \times \beta + \epsilon) \subseteq \cup_{\delta<O}(\alpha \times \delta)$. Thus $\cup_{\delta<O}(\alpha \times \delta) = \cup_{\epsilon<J}(\alpha \times \beta + \epsilon)$, completing the transfinite induction that ordinal multiplication is left distributive. □

(c) We prove this by transfinite induction on γ. The case for 0 holds, as $\alpha \times (\beta \times 0) = \alpha \times 0$ and $(\alpha \times \beta) \times 0 = 0$. Suppose the result holds for some γ. Then

$$\alpha \times (\beta \times \gamma^+) = \alpha \times ((\beta \times \gamma) + \beta)$$

$$= \alpha \times (\beta \times \gamma) + \alpha \times \beta \text{ (Left distributivity)}$$

$$= (\alpha \times \beta) \times \gamma + \alpha \times \beta \text{ (Induction assumption)}$$

and

$$(\alpha \times \beta) \times \gamma^+ = (\alpha \times \beta) \times \gamma + \alpha \times \beta$$

That is, $\alpha \times (\beta \times \gamma^+) = (\alpha \times \beta) \times \gamma^+$.

On the other hand, suppose that the result holds for all ordinals less than some limit ordinal λ. Then $\alpha \times (\beta \times \lambda) = \alpha \times \cup_{\gamma<\lambda}(\beta \times \gamma)$ and $(\alpha \times \beta) \times \lambda = \cup_{\gamma<\lambda}((\alpha \times \beta) \times \gamma)$. Let $O = \cup_{\gamma<\lambda}(\beta \times \gamma)$. Because λ is a limit ordinal, O is also a limit ordinal, meaning $\alpha \times O = \cup_{\delta<O}(\alpha \times \delta)$. We show that $\alpha \times O = \cup_{\gamma<\lambda}((\alpha \times \beta) \times \gamma)$.

On the one hand, suppose that $x \in \alpha \times O$. Then $x < \alpha \times \delta$ for some $\delta < O$. By definition of O, this means that $\delta < \beta \times \gamma$ for some $\gamma < \lambda$, meaning $x < \alpha \times \delta < \alpha \times (\beta \times \gamma) = (\alpha \times \beta) \times \gamma$ (by the transfinite induction assumption). Hence $x \in \cup_{\gamma<\lambda}((\alpha \times \beta) \times \gamma)$. Thus $\alpha \times O \subseteq \cup_{\gamma<\lambda}((\alpha \times \beta) \times \gamma)$.

On the other hand, suppose that $x \in \cup_{\gamma<\lambda}((\alpha \times \beta) \times \gamma)$. Then $x < (\alpha \times \beta) \times \gamma$ for some $\gamma < \lambda$. Thus, by the induction assumption, $x < \alpha \times (\beta \times \gamma)$. Now, $\beta \times \gamma < \beta \times \gamma^+$, meaning $\beta \times \gamma < O$, and consequently, $x \in \alpha \times O$. Thus $\cup_{\gamma<\lambda}((\alpha \times \beta) \times \gamma) \subseteq \alpha \times O$, meaning $\cup_{\gamma<\lambda}((\alpha \times \beta) \times \gamma) = \alpha \times O$, therefore completing the transfinite induction.

Exercise 6.3

(a) $\alpha^1 = \alpha^0 \times \alpha = 1 \times \alpha = \alpha$

(b) We prove this by transfinite induction on α. The case for 0 holds, as $1^0 = 1$. Suppose it holds for some α. Then $1^{\alpha^+} = 1^\alpha \times 1 = 1 \times 1 = 1$. Now suppose that for all $\alpha < \lambda$ for a limit ordinal λ, the result holds. Then $1^\lambda = \cup_{\alpha<\lambda}(1^\alpha) = \cup_{\alpha<\lambda} 1 = 1$.

(c) We prove this by transfinite induction on γ. Clearly this holds when $\gamma = 0$. The cases where $\alpha \leq 1$ are obvious, so we suppose that $\alpha > 1$. Suppose it holds for some γ. Then

$$\alpha^{\beta+\gamma^+} = \alpha^{(\beta+\gamma)^+} = \alpha^{\beta+\gamma} \times \alpha$$

$$= (\alpha^\beta \times \alpha^\gamma) \times \alpha$$

$$= \alpha^\beta \times (\alpha^\gamma \times \alpha)$$

$$= \alpha^\beta \times \alpha^{\gamma^+}$$

On the other hand, suppose that it holds for all ordinals less than some limit ordinal λ. Then

$$\alpha^{\beta+\lambda} = \alpha^{\cup_{\gamma<\lambda}(\beta+\gamma)}$$

and

$$\alpha^{\beta} \times \alpha^{\lambda} = \alpha^{\beta} \times \cup_{\gamma<\lambda}(\alpha^{\gamma})$$

Let $O = \cup_{\gamma<\lambda}(\beta+\gamma)$. Note that O is a limit ordinal. Thus $\alpha^{\beta+\lambda} = \alpha^{O} = \cup_{\gamma<O}(\alpha^{\gamma})$. Let $J = \cup_{\gamma<\lambda}(\alpha^{\gamma})$, which is also a limit ordinal. Thus $\alpha^{\beta} \times J = \cup_{\gamma<J}(\alpha^{\beta} \times \gamma)$. We aim to show that $\cup_{\gamma<O}(\alpha^{\gamma}) = \cup_{\gamma<J}(\alpha^{\beta} \times \gamma)$.

On the one hand, suppose that $x \in \cup_{\gamma<O}(\alpha^{\gamma})$. Then, for some $\gamma < O$, $x < \alpha^{\gamma}$. By definition of O, for some $\delta < \lambda$, $\gamma < \beta+\delta$, meaning $x < \alpha^{\gamma} < \alpha^{\beta+\delta} = \alpha^{\beta} \times \alpha^{\delta}$ (by the induction assumption). Then, by the definition of J, $x \in \cup_{\gamma<J}(\alpha^{\beta} \times \gamma)$. Thus $\cup_{\gamma<O}(\alpha^{\gamma}) \subseteq \cup_{\gamma<J}(\alpha^{\beta} \times \gamma)$.

On the other hand, suppose that $x \in \cup_{\gamma<J}(\alpha^{\beta} \times \gamma)$. Then, for some $\gamma < J$, $x < \alpha^{\beta} \times \gamma$. By the definition of J, $\gamma < \alpha^{\delta}$ for some $\delta < \lambda$. Hence $x < \alpha^{\beta} \times \gamma < \alpha^{\beta} \times \alpha^{\delta} = \alpha^{\beta+\delta}$, and by definition of $\cup_{\gamma<O}(\alpha^{\gamma})$, it follows that $x \in \cup_{\gamma<O}(\alpha^{\gamma})$. Thus $\cup_{\gamma<J}(\alpha^{\beta} \times \gamma) \subseteq \cup_{\gamma<O}(\alpha^{\gamma})$, and consequently, $\cup_{\gamma<J}(\alpha^{\beta} \times \gamma) = \cup_{\gamma<O}(\alpha^{\gamma})$. Thus the transfinite induction is complete.